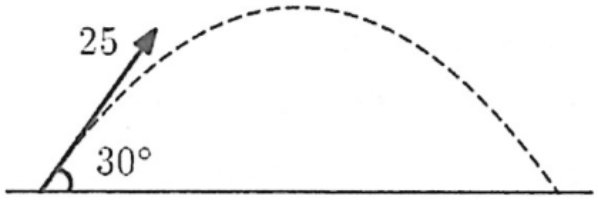
Projectiles

Questions 1–20

Question 1. The following questions use $g = 9.8$ ms^{-2} A particle is projected from point O on horizontal ground with a velocity of 25 ms^{-1} at an angle of 30° above the horizontal before hitting the ground.



Find

a) the velocity of the particle after 0.5 seconds.
b) the greatest height the particle reaches.
c) the time the particle spends in the air.
d) the total horizontal distance travelled by the particle.

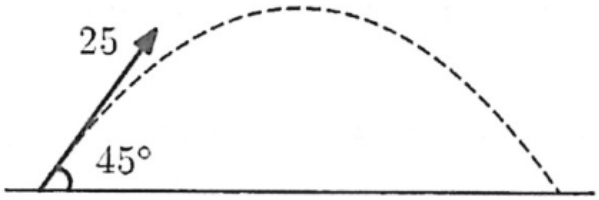
Question 2. A particle is projected from point O on horizontal ground with a velocity of 10 ms^{-1} at an angle of 30° above the horizontal before hitting the ground.



Find

a) the velocity of the particle after 0.4 seconds.
b) the greatest height the particle reaches.
c) the time the particle spends in the air.
d) the total horizontal distance travelled by the particle.

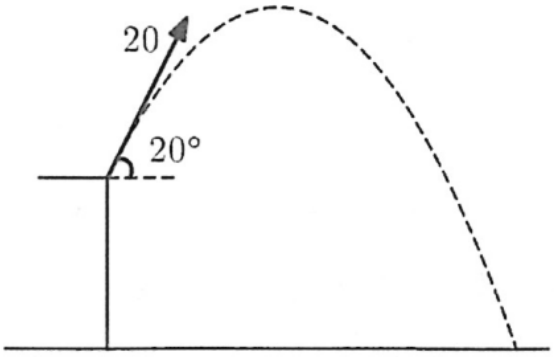
Question 3. A particle is projected from point O on horizontal ground with a velocity of 25 ms^{-1} at an angle of $45°$ above the horizontal before hitting the ground.



Find

a) the velocity of the particle after 3 seconds.
b) the greatest height the particle reaches.
c) the time the particle spends in the air.
d) the total horizontal distance travelled by the particle.

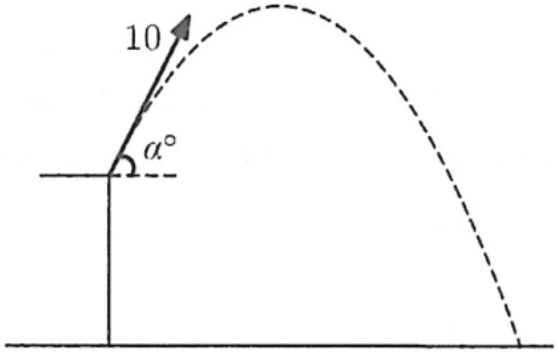
Question 4. A ball is projected at an angle of $20°$ above the horizontal from a tower of 10 metres high at a speed of 20 ms^{-1} before landing on the floor.



Modelling the ball as a particle, find

a) the initial velocity of the ball.
b) the time the ball spends in the air.
c) the velocity of the ball as it hits the ground.

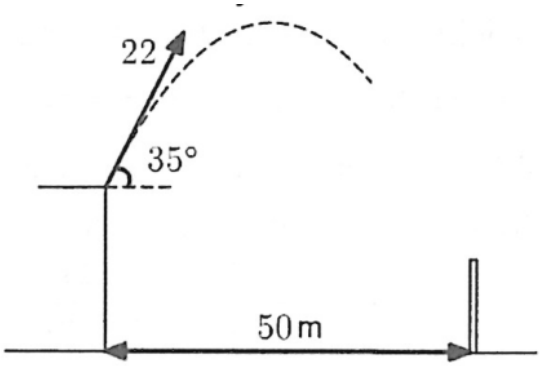
Question 5. A stone is thrown from a cliff of height 80m into a lake below. It is projected at 10 ms^{-1} at an angle of α above the horizontal where $\tan\alpha = \frac{4}{3}$



By modelling the stone as a particle, find

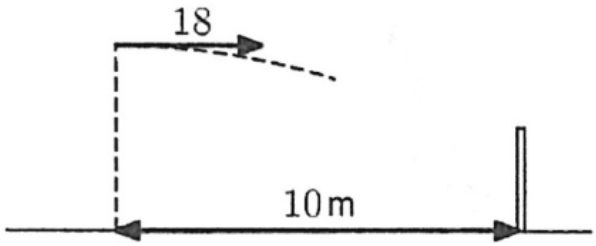
a) the time the stone spends in the air
b) the vertical and horizontal components of the displacement of the stone in terms of t and hence show that the equation of the trajectory of the stone is $y = 80 + \frac{4}{3}x - \frac{49}{360}x^2$
c) the maximum height of the stone above the ground.

Question 6. A ball is projected at an angle of 35° above the horizontal from a height of 4.5 metres and at a speed of 22 ms^{-1}. A wall of height 1.5 metres stands 50 metres away.



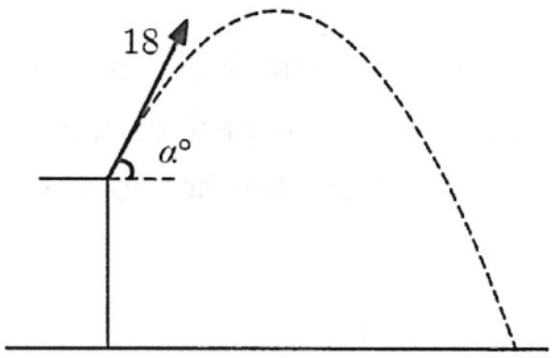
Does the ball hit the wall?

Question 7. A tennis ball is hit horizontally with a speed of 18 ms^{-1} from a height of 2.5 metres. The net is 1.2 metres high and is 10 metres away.



a) Will the ball pass over the net?
b) What modelling assumptions have you made?

Question 8. A stone is thrown at speed 18 ms^{-1} at an angle of $\alpha°$ where $\tan\alpha = \frac{5}{12}$ above the horizontal from a cliff of height 25 metres.



Modelling the stone as a particle, find the amount of time that the ball is more than 27 metres above ground level.

Question 9. A stone is thrown at a speed of u ms^{-1} with angle of α above the horizontal where $\tan\alpha = \frac{3}{4}$ from a cliff of height 40 m. It travels 50 metres horizontally before hitting the water below t seconds later.

a) Find the exact value of t

b) Find the exact value of u

c) State any modelling assumptions you have made.

A stone of twice the mass is thrown from the cliff, otherwise the conditions remain the same.

d) Explain what would change the results in the previous part of the question.

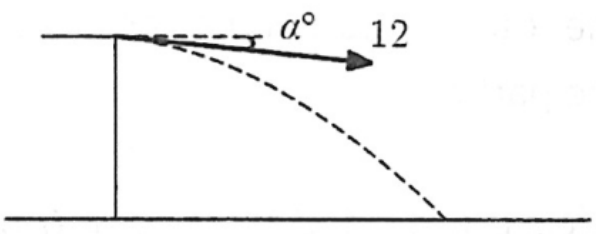
Question 10. A particle is thrown at 10 ms^{-1} at an angle of α below the horizontal, where $\tan\alpha = \frac{2}{5}$, from a height of 2 metres. Show that the equation of the trajectory of the particle is

$$y = 2 - \frac{2}{5}x - \frac{1421}{25000}x^2$$

Question 11. The following questions use $g = 9.81\ \text{ms}^{-2}$ Dean kicked a football on a horizontal field with speed $20\ \text{ms}^{-1}$ at an angle of θ above the horizontal where $\tan\theta = \frac{3}{4}$. The football goal is 36 metres away, the top of the goal post is 2.4 metres above the ground.

a) Does the ball go into the goal?
b) State any modelling assumptions you have made.

Question 12. A stone is thrown towards some water at speed $12\ \text{ms}^{-1}$ at an angle of $\alpha°$ below the horizontal where $\tan\alpha = \frac{1}{10}$



It lands on the surface of the water 0.4 seconds later. By modelling the stone as a particle, find

a) the direction of the motion when 0.2 seconds have passed.
b) the height above the water surface area that the stone was thrown.
c) the horizontal distance the stone travelled before hitting the water.

Question 13. The following questions use $g = 10$ ms^{-2} A ball is projected at an angle of $\theta°$ above the horizontal from a height of 2.5 metres above the ground. The ball hits a target which is 1.5 metres above the ground and 7 metres away from the point of projection, 2 seconds after projection. Modelling the ball as a particle, find

a) the speed at which the ball is projected.
b) the angle at which the ball is projected.

Question 14. A ball is projected at angle of 30° above the horizontal from a height of 2 metres at a speed of 18 ms^{-1}. It passes over a wall of height 3 metres stands d metres away. Find

a) the maximum value of d such that the ball still passes over the wall.
b) the direction at which the ball impacts the ground.

Question 15. Two balls A and B are projected from a bridge at the same point. Ball A is projected with speed 10 ms^{-1} at an angle of 30° to the horizontal. Ball B is projected 1 second after ball A with speed 40 ms^{-1} at an angle of 60°. By modelling the balls as particles, identify the vertical distance between the balls when one passes over the other, assuming they have not reached the ground.

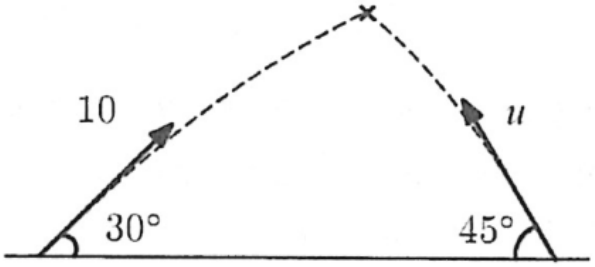
Question 16. Prerequisite knowledge for Q16: Reciprocal Trigonometric Functions A child shoots a water pistol, with the water firing at 10 ms^{-1} at an acute angle of θ° above the horizontal.

a) Show that the equation of the trajectory of the water fired from a height of a metres above the ground is $y = a + x\tan\theta - \frac{x^2}{20\cos^2\theta}$

The water is fired from a height of 0.7 metres above the ground and reaches a horizontal distance of 9.2 metres from the child.

b) Find the angle(s) above the horizontal at which the water pistol was fired.

Question 17. The following questions use exact values of g Two balls A and B are projected at the same time towards each other from points P and Q respectively, where P and Q lie 5 metres apart.
Ball A is projected with speed 10 ms^{-1} at 30° to the horizontal.
Ball B is projected with speed u ms^{-1} at 45° to the horizontal.



The two balls collide.

a) Show that the value of u is $5\sqrt{2}$
b) Find the exact time at which they collide giving your answer to 2 significant figures.
c) Find the horizontal distance ball A is from its start point when they collide, giving your answer to 3 significant figures.

Question 18. A particle is projected with speed u ms^{-1} from the ground at an angle of θ above the horizontal. Show that the equation of the trajectory of the particle is

$$y = \tan\theta\, x - \frac{g}{2u^2}\sec^2\theta\, x^2$$

Question 19. A particle is projected with speed u ms^{-1} at an angle of θ above the horizontal.
The particle reaches a maximum vertical height of h metres.

a) Show that $h = \frac{u^2}{2g}\sin^2\theta$

It is known that $u = 10$ ms^{-1}.

b) Given θ is acute show that $0 \leq h \leq \frac{50}{g}$

Given that a projectile needs to have a maximum height of 5 metres above its start point and is projected at 12 ms^{-1}.
Using $g = 10$ ms^{-2},

c) find the acute angle θ at which it must be projected, giving your answer to 1 decimal place.

Question 20. Prerequisite knowledge for Q20: Double Angle Formulae A particle is projected from the ground with speed u ms^{-1} at an angle of $\theta°$ above the horizontal.
The particle lands x metres from its original start location after t seconds have passed.

a) Show that $t = \frac{2u}{g}\sin\theta$

b) Show that $x = \frac{u^2}{g}\sin 2\theta$

It is known that the projectile needs to land at least 20 metres away from the start point and is projected at 15 ms^{-1}.
Using $g = 10$ ms^{-2},

c) find the range of possible angles θ at which it must be projected, giving your answers to 1 decimal place.